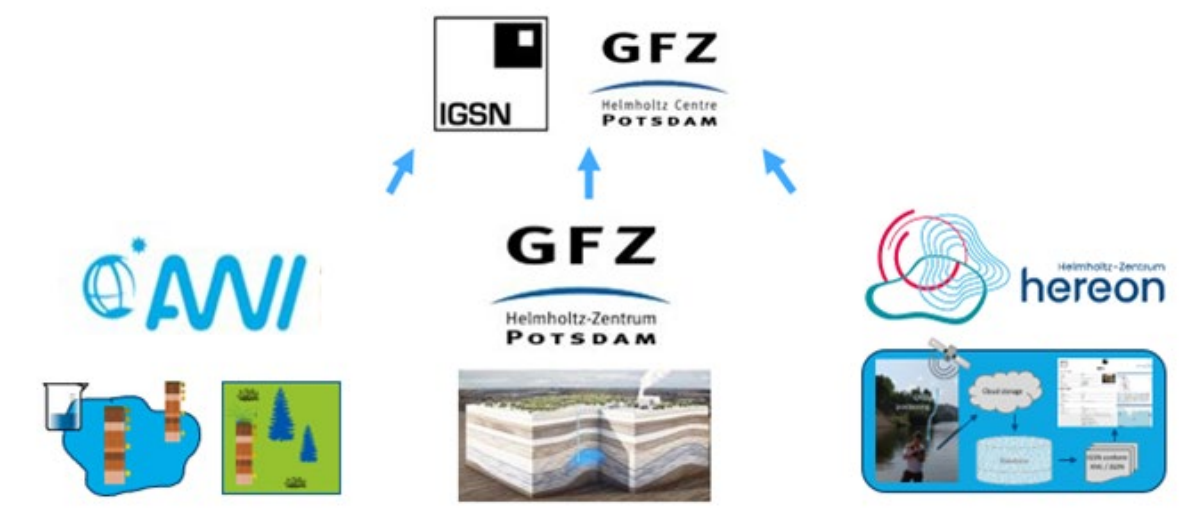


FAIR Workflows to establish IGSN for Samples in the Helmholtz Association

D7 – Standard Operating Procedure for automatic IGSN registration

(Hereon, GFZ, AWI)



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1. Introduction

1.1. Purpose of this document

This document contains deliverable D7 *Standard Operating Procedure for automatic IGSN registration* of the project **FAIR Workflows** to establish IGSN for **S**amples in the Helmholtz Association (FAIR WISH) funded by the Helmholtz Metadata Collaboration (HMC). Deliverable D7 is part of work package 6 *IGSN registration workflows for biogeochemical sample databases*.

This deliverable builds upon D2 *Exemplary standardised metadata templates for Geo-Bio Samples (vegetation, water, sediment, rock samples) for all uses cases* (Wieczorek et al., 2022), the FAIR WISH “FAIR SAMPLES Template” (Wieczorek et al., 2023) and D5 *Mapping of database metadata to machine readable IGSN metadata (Use Case Hereon)* (Baldewein et al., 2023) to register IGSNs for a subset of samples stored in the Hereon biogeochemical campaign database.

1.2. Intended users

This standard operating procedure is intended for data managers that want to automate the registration process of IGSNs based on information stored in relational databases. It is not intended for individuals who want to make a one-time submission based on the FAIR SAMPLES Template.

2. Standard operating procedure

2.1. Prepare FAIR WISH Sample description template

2.1.1 Download the newest version of the FAIR SAMPLES Template at <https://doi.org/10.5281/zenodo.7520015>

2.1.2 Use the ‘Guide’ of the template (second page in the spreadsheet) to find all potentially relevant metadata fields for your samples. It is easier to discard variables at the end, if not needed, rather than adding fields later in the mapping process. You will need to have in depth knowledge of your samples for this step.

2.1.3 Identify all controlled vocabulary lists that will be needed for the selected fields.

2.2. Map metadata to IGSN fields

2.2.1 Initially separate all samples based on their hierarchical relationship in parents and children and treat them independently, as they most likely will have varying metadata. From now on, ensure that you repeat each step for each hierarchy level.

2.2.2 Create a mapping table containing the columns “IGSN field name”, “Variability” and “Data”. Fill all IGSN fields identified in step 2.1.2 into the column “IGSN field name”.

2.2.3 For each IGSN field name, decide if the content of the field is constant for all samples or dependent on a database field. Assign a value of “variable” or “constant” to each field based on this assessment.

2.2.4 For all constant fields, write the constant value in the column “Data”. This may for example be the location of the sample archive.

2.2.5 For variable fields, find the corresponding table and column in your database and write it in the column “Data”.

2.3. Map data to controlled vocabulary lists

2.3.1 Based on the controlled vocabulary lists identified in 2.1.3, check your corresponding data based on the information in your mapping table. If your data does not adhere to the controlled vocabulary, map your data to the vocabulary, ideally in a new database table.

2.3.2 Correct the used data column in the mapping table to your new database table.

2.3. Write IGSN metadata

2.3.1 Using your favorite programming language, extract all relevant metadata information for the samples that you want to assign IGSNs to based on the “data” column in your mapping table.

2.3.2 Write a csv file with your IGSN field names as headings.

2.3.3 For each sample, fill in all IGSN field names based on your database content or the previously defined constant values.

Step 2.3 is the first fully automatable step. The previous steps will need to be carried out manually once.

2.4. Register IGSNs

Send your CSV file containing the IGSNs to your registration authority via e-mail. A new application programming interface (API) for automatisisation of this process is currently in development by GFZ Data Services. This was necessary with the new IGSN DataCite partnership which included the change of persistent identifier type for IGSN from Handles to DataCite DOI.

3. References

Baldewein, Linda, Leefmann, Tim, Kleeberg, Ulrike, Brauser, Alexander, Elger, Kirsten, Frenzel, Simone, Heim, Birgit, & Wieczorek, Mareike. (2023). FAIR WISH D5 – Mapping of database metadata to machine readable IGSN metadata (Use Case Hereon). Zenodo. <https://doi.org/10.5281/zenodo.7788654>

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4. Appendix

4.1. List of acronyms

AWI	Alfred Wegener Institute for Polar and Marine Research
FAIR	Guiding Principles for Findable, Accessible, Interoperable and Reusable research data (Wilkinson et al., 2015, https://doi.org/10.1038/sdata.2016.18)
FAIR WISH	FAIR Workflows to establish IGSN for Samples in the Helmholtz Association
GFZ	German Research Centre for Geosciences
IGSN	International Generic Sample Number

4.2. Example mapping table

IGSN field name	Variability	Data
igsn	variable	sample.IGSN
parent_igsn	variable	station.IGSN
name	variable	sample.SAMPLE_NAME
sample_other_names	variable	sample.SAMPLE_LABEL
collection_start_date	variable	sample.DATA_DATETIME
collection_end_date	variable	sample.DATA_DATETIME
current_archive	constant	Your institution
current_archive_contact	constant	Contact person
sampleAccess	constant	Samples are not stored but destroyed after measurements.